ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2
PAPER 2 November 2020
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Mathematical tables or a non-programmable calculator may be used.
INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer all questions

1. (a) Expand and simplify $(2x - 2)(x + 2)$. [3]
(b) Factorise $x^2 + 9x + 20$. [2]
(c) Solve $(x + 2)/2 = 4$. [3]
(d) Simplify $(x^2 - 16)/(x - 4)$. [2] **[10]**

2. Lines AB and CD are parallel. Transversal EF meets AB at P and CD at Q.
(a) Angle APQ = 52°. State the corresponding angle at Q and give a reason. [2]
(b) Find the co-interior angle to 52° and give a reason. [2]
(c) A triangle has angles 52°, 37° and x°. Find x. [2]
(d) Is the triangle acute, right-angled or obtuse? Give a reason. [2] **[8]**

3. A rectangular garden in Chitungwiza measures 12 m by 11 m. A path of width 2 m runs all around the outside.
(a) Find the area of the garden. [2]
(b) Find the area of the path. [3]
(c) A circular flower bed of radius 7 m is later dug inside the garden. Find its area. Take $\pi = 22/7$. [3] **[8]**

4. The line L has equation $y = 1x + 1$.
(a) State the gradient and the y-intercept. [2]
(b) Find the coordinates of the point where L meets the x-axis. [2]
(c) Point A is (0, 1). Show that A lies on L. [2]
(d) Write the equation of a line parallel to L that passes through (0, 4). [2] **[8]**

5. The marks of 31 Form 4 pupils on a 6-mark quiz were:
3 (5 pupils), 2 (8 pupils), 5 (8 pupils), 4 (7 pupils), 6 (3 pupils).
(a) Find the modal mark. [1]
(b) Find the mean mark. [4]
(c) Find the range. [2]
(d) One extra pupil scores 6. Describe the effect on the mean. [2] **[9]**

6. A solar pump is marked at \$800 in Harare.
(a) A school gets a 10% discount. Calculate the discounted price. [3]
(b) VAT of 15% is then added. Calculate the final amount paid. [3]
(c) The school pays a deposit of 40% of the final amount. How much remains to be paid? [3] **[9]**

SECTION B (48 marks) — Answer any four questions

7. (a) Solve $x^2 - 7x + 10 = 0$ by factorisation. [4]
(b) The length of a rectangular maize plot is 5 m more than the width. The area is 24 m^2? Using width w, form an equation and solve to find possible dimensions, taking the quadratic $(w + 2)(w + 5)$ is NOT required — instead solve $w(w + 5) = 14$. [5]
(c) Sketch $y = (x - 2)(x - 5)$, showing intercepts. [3] **[12]**

8. A tree at Heroes Acre is observed from point A on level ground. The angle of elevation of the top is 30°. Distance from A to the foot of the tree is 24 m.
(a) Calculate the height of the tree. [3]
(b) From B, 24 m due East of A, the bearing of the tree foot is 300°. Sketch and mark the information. [3]
(c) Use $\tan 30° = 1/\sqrt{3}$ or $\sin 30° = 1/2$ as appropriate. Leave √ in the answer if needed. [3]
(d) A bird sits halfway up the tree. Find the angle of elevation from A to the bird. [3] **[12]**

9. O is the centre of a circle. A, B and C are points on the circumference. Angle BAC = 44° and AB is a diameter.
(a) Find angle ACB. Give a reason. [3]
(b) Find angle BOC (angle at the centre standing on arc BC). [3]
(c) If D is on the remaining circumference, find angle BDC. Reason. [3]
(d) State the theorem: angle in a semicircle. [3] **[12]**

10. Position vector OA = a = (3 ; 2), OB = b = (1 ; 6). M is the midpoint of AB.
(a) Find vector AB. [3]
(b) Find |AB|. [3]
(c) Find position vector OM. [3]
(d) Show that AM = MB. [3] **[12]**

11. 50 pupils sat a Mathematics test (marks 0–50). The cumulative frequency curve (ogive) is drawn.
(a) Explain how to read the median from the ogive. [3]
(b) The median is estimated as 32. What does this mean for a typical pupil? [2]
(c) Describe how to find the interquartile range from the graph. [4]
(d) Why is the IQR useful compared with the range? [3] **[12]**

12. Triangle P(1, 1), Q(3, 1), R(1, 4) is drawn on a grid.
(a) Draw the image P'Q'R' after a reflection in the y-axis. State the coordinates of P'. [4]
(b) Translate PQR by vector (2 ; −1). Give Q''. [3]
(c) The matrix (0 −1 ; 1 0) represents a transformation. Describe it fully and find the image of (3, 1). [5] **[12]**

13. (a) y varies directly as x. When x = 4, y = 36. Find y when x = 10. [4]
(b) z varies inversely as x. When x = 3, z = 8. Find z when x = 6. [4]
(c) A farmer has at most 40 hours. Planting maize takes 2 hours per hectare and vegetables 1 hour. Write an inequality for hectares m and v. [4] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).